PH 724 Data Analysis & Scientific Communication Incorporating Use of Generative AI

Part 7: Introduction to the Gaussian Process

Zhiyang Zhou (zhou67@uwm.edu, zhiyanggeezhou.github.io)

2026/10/18 23:18:13

What is a Gaussian process (GP)

- **Intuition:** a series of random variables are connected to each other, forming a random curve.
- Any finite set of points on the random curve has a joint multivariate normal distribution.
- Specifically, in math, we write a random curve $Y(t) \sim \text{GP}\{\mu(t), K(t,t')\}$ if, for any finite collection of times $t_1, \ldots, t_m$,

$$\begin{pmatrix} Y(t_1) \\ \vdots \\ Y(t_m) \end{pmatrix} \sim N\left(\begin{pmatrix} \mu(t_1) \\ \vdots \\ \mu(t_m) \end{pmatrix}, [K(t_j, t_k)]_{j,k=1}^{m} \right).$$

Two ingredients: mean and covariance functions

- Mean function: $\mu(t) = E\{Y(t)\}$
  - Describes the expected trajectory
- Covariance function: $K(t,t') = \text{Cov}\{Y(t), Y(t')\}$
  - Describes how the covariance varies across times.

The covariance function matters

Consider two possible covariance structures:

- **Smooth covariance:** nearby times tend to have similar deviations; sample paths are smooth.
- **Brownian-motion covariance:** deviations accumulate over time; paths are continuous but rough.

Sample paths of a GP with a smooth covariance function

```
library(ggplot2)
library(MASS)

set.seed(724)

# Number of simulated curves
n_curves <- 30

t <- seq(0, 1, length.out = 100)
```

```r
mu <- 30 * t^2 / (0.25 + t^2)

K <- outer(t, t, function(a, b)
  6^2 * exp(-(a-b)^2 / (2*0.18^2)))

# Generate more simulated trajectories
paths <- MASS::mvrnorm(n_curves, mu, K + diag(1e-7, length(t)))

# Create data frame
d <- data.frame(
  time = rep(t, n_curves + 1),
  height = c(mu, as.vector(t(paths))),
  trajectory = rep(
    c("Mean", paste("Possible plant", 1:n_curves)),
    each = length(t)
  )
)

# Plot
ggplot(d, aes(time, height,
              group = trajectory,
              color = trajectory == "Mean")) +
  geom_line(aes(alpha = trajectory == "Mean"), linewidth = 0.8) +
  scale_color_manual(
    values = c("TRUE" = "#202020", "FALSE" = "#7a9fbd"),
    guide = "none"
  ) +
  scale_alpha_manual(
    values = c("TRUE" = 1, "FALSE" = 0.4),
    guide = "none"
  ) +
  labs(
    x = "Scaled time",
    y = "Height (illustrative units)",
    subtitle = "Illustration only: 30 simulated trajectories from a smooth GP"
  ) +
  theme_minimal(base_size = 16)
```

Brownian motion as a Gaussian process

For standard Brownian motion $W(t)$, starting at $W(0) = 0$,

$$E\{W(t)\} = 0, \qquad \text{Cov}\{W(t), W(t')\} = \min(t, t').$$

- The start point of each trajectory is fixed at 0. - Variability increases with elapsed time: $\text{Var}\{W(t)\} = t$. - Brownian paths are “bumpy”.

Sample paths of a Brownian motion

```r
library(ggplot2)

set.seed(724)
```

```r
# Number of simulated Brownian-motion paths
n_curves <- 30

tb <- seq(0, 1, length.out = 150)

# Generate Brownian-motion paths
B <- replicate(
  n_curves,
  c(0, cumsum(rnorm(length(tb) - 1, sd = sqrt(diff(tb)[1]))))
)

# Create data frame
db <- data.frame(
  time = rep(tb, n_curves),
  deviation = as.vector(B),
  plant = rep(1:n_curves, each = length(tb))
)

# Plot
ggplot(db, aes(time, deviation, group = plant)) +
  geom_line(
    color = "#7a9fbd",
    linewidth = 0.8,
    alpha = 0.4
  ) +
  geom_hline(
    yintercept = 0,
    color = "#202020",
    linewidth = 1,
    linetype = "dashed"
  ) +
  labs(
    x = "Elapsed time",
    y = "Deviation",
    subtitle = "Illustration only: 30 simulated standard Brownian-motion paths"
  ) +
  theme_minimal(base_size = 16)
```

Real-world applications of GPs

- One of the most influential mathematical finance models is the Black–Scholes model, developed in 1973. It assumes that the stock price follows a geometric Brownian motion (whose logarithm is a GP). Black-Scholes uses this stochastic model primarily for option pricing, not for reliably predicting tomorrow's stock price.
- Google has used GP-based Bayesian optimization to efficiently tune machine-learning models and other complex systems. GPs help identify promising configurations while reducing the number of expensive evaluations. See https://deepmind.google/research/publications/108347/.
- GPs are widely used to model spatially varying quantities, such as environmental pollution levels and disease prevalence.
  - For example, the Malaria Atlas Project uses GP-based spatial modeling to estimate malaria prevalence in regions with limited observational data. Its broader modeling efforts also estimate malaria incidence and mortality, producing high-resolution maps that support public health

research and decision-making.